

is functional for 30 days, but with a few restrictions. It has no support for MISRA and source code for runtime libraries is not included. The evaluation version also has restriction on code size. It would be great if all the features could be provided to the developers.

Home Page: <http://www.iar.com/web-site1/1.0.1.0/3/1/>

Download URL: <http://www.iar.com/web-site1/1.0.1.0/675/1/>

Platform Supported: Windows

Microprocessor Simulator

Microprocessor Simulator is for learners in the 16+ age group. The instruction set of this software is similar to Intel 8086. The latest version of the software is 5.0.

What's Special?

This simulator contains a small subset of the full instruction set normally found in any processor. It includes advanced instructions, including interrupts. It emulates an 8-bit CPU having 16 input and output ports. It also contains an assembler and an interrupt, which is triggered from hardware. The best part is that the CPU clock speed can be altered. The peripherals that can be connected to the simulator are keyboards, traffic lights, seven-segment displays, heaters and thermostats, stepper motors, etc.

Neil Bauers, the developer, who works at East Norfolk Sixth Form College as an electronics and computing teacher, says, "The program has evolved over years and reached maturity. It's possible to write some very simple beginners' programs as well as use advanced techniques like interrupt processing."

Users' feedback

The software gives the programmers a real-time view of the RAM and processor registers while the code is running. The programmer can 'single step' through the lines of code or run programs continuously at a range of speeds. The docu-

Embedded software inside EFY-CD

- PIC Development Studio for PIC 16F84 – March 2008 and February 2010
- WinAVR – June 2008 and March 2010
- IAR Embedded Workbench for TI TSP430 architecture – July 2008
- IAR visualSTATE – July 2009
- GPSIM – February 2010
- ICPROG – February 2010
- mikroC – February 2010

mentation includes lots of examples.

Suggestion

Neil says, "The program is based on the legacy Intel 8086/8088 family of chips. A similar simulator for the ARM processor might be more useful in the 21st century. Also the machine language is invented and no hardware exists to run the code."

Home Page: <http://www.softwareforeducation.com/>

Download URL: <http://www.softwareforeducation.com/sms32v50/>

Platform Supported: Windows

WinAVR

This open source software platform for Atmel's AVR series of RISC microprocessors works under the GNU GPL licence. Its latest version was released on January 10, 2010.

What's Special?

This open source tool is useful for everyone, starting from hobbyists and school students to commercial project developers. It is basically a collection of tools that comprises a compiler, a programmer and a debugger. Further, it has AVARIC, which connects the JTAG ICE to the AVR-GDB debugging software, a makefile editor called 'MFile', SRecord tool for editing EPROM load files and much more.

Users' feedback

WinAVR is very easy to use. It has good support for both users and de-

velopers.

Suggestion

The software was released just a month ago, so till date no bugs have been filed for it. But, the current WinAVR bugs may be found at http://sourceforge.net/tracker/?group_id=68108&atid=520074

Home Page: <http://winavr.sourceforge.net/>

Download URL: <http://winavr.sourceforge.net/download.html>

Platform Supported: Windows

Oshonsoft 8085

OshonSoft 8085 simulator IDE for Intel 8085 processor comprises a compiler, an assembler, a disassembler and a debugger. It was developed by Vladimir Soso.

What's Special?

The software is for everyone, educators as well as developers. It comes with a simulation interface that shows the internal processor architecture with interrupts interface. For simulation of input/output (I/O) instructions, it comes with an I/O ports editor and peripheral devices interface. Further, it comes with variable simulation rate, simulation statistics and simulation logging with log viewer. The debugger gives you the option to create break points for step-by-step debugging.

Users' feedback

The debugger has helped many developers to write their code perfectly and later burn it in the EPROM. It is a great tool for beginners, as it is easy to use and shows the dynamic variable changes happening in the memory when the simulator is running.

Suggestion

It would be great if the evaluation version was available longer.

Home Page: <http://www.oshonsoft.com/>

Download URL: <http://www.oshonsoft.com/8085.html>

Platform Supported: Windows ●

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